HEYMAX

Take-Home Assessment

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- GitHub Repository
- Tableau Public

Overview

Part 1 - Analytics Stack

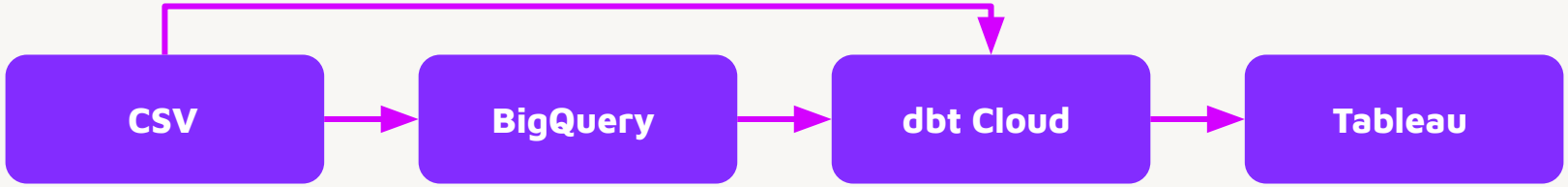
- dbt data model
- Interactive dashboard (Tableau)
- Limitations

Part 2 - AI Agent Design

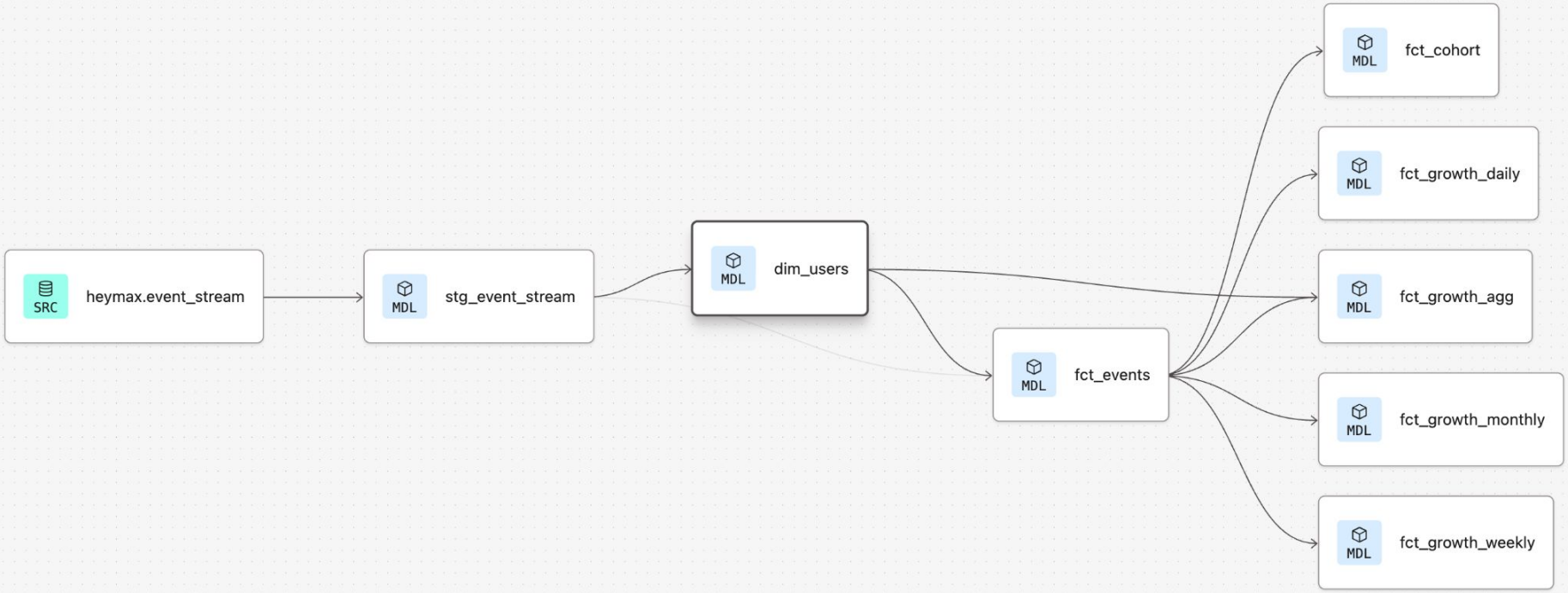
- Architecture Design
- Human in the loop
- Failure Modes & Observability
- Scope & Build Plan
- Limitations & Trade offs

End-to-End Analytics Stack

Analytics Stack



Data Model Lineage



Dashboard - 1

Period & Country filter

KPI cards anchor to latest full period (Nov 2025 — Dec is partial)

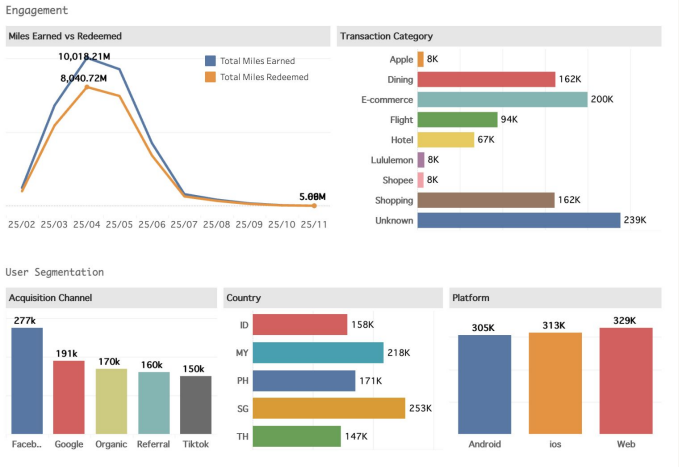
Churned bars below x-axis

Quick Ratio for acquisition signals.

Dashboard - 2

Cohort Retention - Consecutive

	0	1	2	3	4	5
2025/02	100.0%	100.0%	100.0%	100.0%	100.0%	94.0%
2025/03	100.0%	42.7%	24.7%	11.3%		
2025/04	100.0%	43.1%	23.9%	11.5%		
2025/05	100.0%	43.8%	24.1%	11.5%		
2025/06	100.0%	42.5%	23.3%	9.9%		
2025/07	100.0%	37.3%	9.4%			
2025/08	100.0%	37.5%	12.3%			
2025/09	100.0%	23.6%	1.4%			
2025/10	100.0%	8.8%				
2025/11	100.0%					



Consecutive cohort, resets on churn

Engagement and User Segmentation

What the Build Taught Me

No.	Taught Items	How if it's implemented	The solution
1.	One model for all grains	- A bug in one grain would silently affect all three. If the monthly classification logic breaks, daily and weekly break with it. - Incremental loading doesn't work cleanly across grains in one table.	- Separate models → failures are isolated. Fails independently, and incremental logic stays clean
2.	NOT EXISTS function for classifications	Works but it rescans the entire table for every row.	LEAD window function → LEAD looks ahead in an already-sorted window.

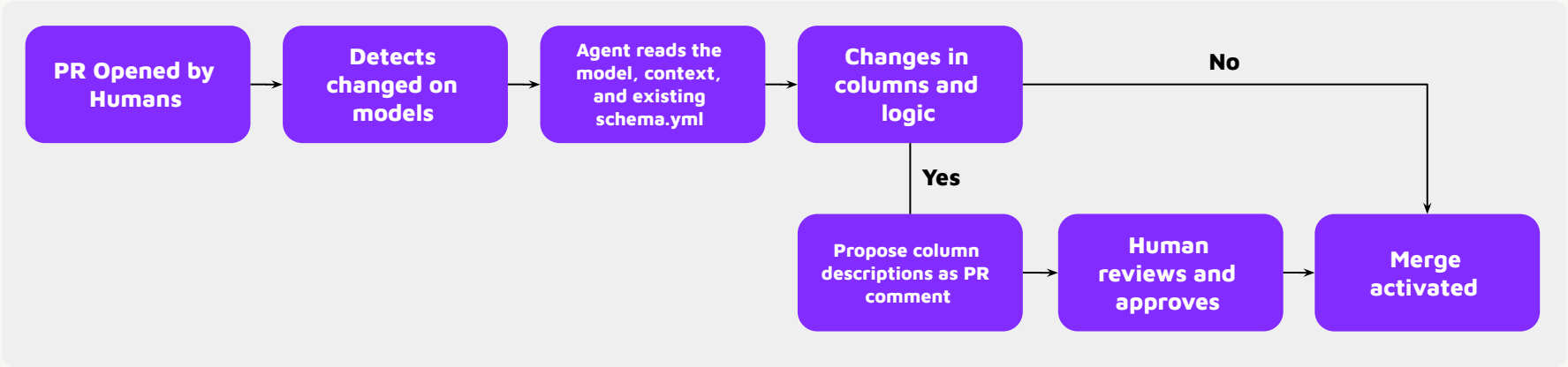
If I have more time & Access

- Add a streaming ingestion layer to replace daily batch
- Implement dbt snapshots for true SCD-2 user history
- Add a logging framework to track model run times and row counts
- Extend CI to run full dbt test on every PR with BigQuery service account auth

AI Agent Design -

Data Documentation Agent

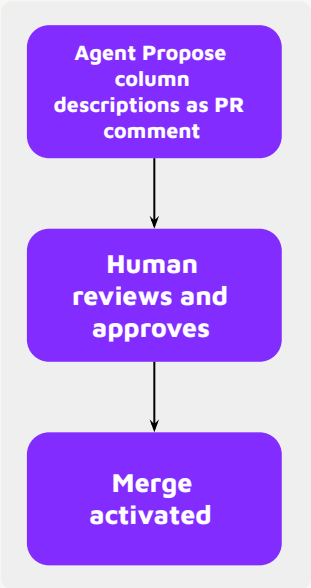
Architecture



The agent compares the current schema.yml and logic against the updated file to identify the changes. Only the changed columns will get a new proposed description. If there's no existing schema.yml the agent will generates the description for all columns.

Human-in-the-Loop Design

The agent is not a decider, it's only propose. Every new documentation changes proposed, it requires human approval before merged.



Preventing Silent Bad Docs

- The agent instructed to create a short and accurate descriptions
- The proposed will shown as a diff against the existing docs, to flag the changes
- The agent flags the columns where the description is uncertain for the ambiguous column name

Surfacing Changes

- PR comment on GitHub
- Slack notification

Failure Modes & Observability

Mode 1

Wrong Column Descriptions

The agent misunderstands the columns means and generate the wrong description.

Mode 2

Agent Skips Changed Columns

The agent misses a newly added columns because of the fails without warning.

Mode 3

Agent Runs but PR Comment is Never Reviewed

Reviewers 100% trust the agent and approve without review it carefully.

Scope & Build Plan

One Week Scope	
In-scope	**Out-of-scope**
GitHub Actions trigger on PR for changed files LLM call to generate column descriptions for new and changed columns PR comment with proposed YAML files	Auto commit on approval Support for existing models with complex lineage

Measuring the Usefulness

1. Time Saved
2. Adoption
3. Coverage

Known Limitations & Tradeoffs

Limitations

- Only knows what's in the prompt and the SQL
- Hallucinations
- Not fully automated

Trade-offs

- Cost vs coverage
- Automation vs control
- Simplicity vs completeness

Thank you